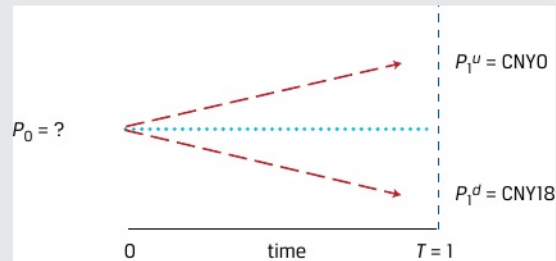


The put option value under the two scenarios is as follows:

- Price Increase (p_1^u): The 40 percent increase results in a put option value of CNY0, as the option owner will choose not to sell the stock at the exercise price.
- Price Decrease (p_1^d): The 20 percent decrease results in a put option value of CNY18 (CNY50 – CNY32), as the put option owner will choose to sell the stock with a market price of CNY32 at the exercise price of CNY50.



Unlike the call option, the put option increases in value as the stock price falls, so a risk-free portfolio combines a stock purchase with a purchased put option. Specifically, assume at $t = 0$, we sell a put option at a price of p_0 and purchase 0.75 units of the underlying stock. Portfolio values at $t = 0$ and $t = 1$ are as follows:

- $V_0 = 0.75S_0 + p_0$,
 - $V_1^u = 0.75S_1^u + p_1^u$, and
 - $V_1^d = 0.75S_1^d + p_1^d$,
- where $V_1^u = V_1^d$.

The value of the risk-free portfolio in one period ($V_1 = 0.75S_1 + p_1$) under the two scenarios is:

- Price Increase (V_1^u): The portfolio value is CNY42 ($= (0.75 \times \text{CNY56}) + \text{CNY0}$).
- Price Decrease (V_1^d): The portfolio value is CNY42 ($= (0.75 \times \text{CNY32}) + \text{CNY18}$).

Solve for p_0 , by setting the present value of the replicating portfolio ($0.75S_0 + p_0$) equal to the discounted future value of the payoff, S_0 is CNY40 and r is 5 percent:

$$V_0 = 0.75S_0 + p_0,$$

$$V_0 = \frac{V_1^u}{1+r} = \frac{V_1^d}{1+r}, \text{ and}$$

$$p_0 = \frac{\text{CNY42}}{1.05} - 0.75(\text{CNY40}).$$

Solve for p_0 to be equal to CNY10. The higher put price is a result of the greater payoff of the put option versus the call option under the given pa-